

AYUSH BHATT

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EDUCATION:

B.Tech in Aerospace Engineering | UPES | Dehradun, India | Jul 2023 – Jun 2027 | CGPA: 7.45/10

School of Excellence, Dwarka Sector 22 | New Delhi, India | Class XII: 82% | Class X: 89%

SKILLS:

Tools: Astrometrica , MATLAB , OpenRocket , OpenFOAM , GMAT , XFLR5

INTERNSHIPS

Student Trainee | ADRDE, DRDO

Agra, India | Jun 2026 – Jul 2026

- Selected for training at Aerial Delivery Research & Development Establishment, DRDO, with exposure to defence aerospace R&D, aerial delivery systems, OpenFOAM-based CFD analysis, technical documentation, and final project reporting.

Summer Intern | SpiceJet

Gurugram, India | Jun 2025 – Jul 2025

- Worked across Engineering and Communications departments, supporting aircraft maintenance documentation, operational reporting, inter-departmental coordination, and aviation workflow understanding.

Social Intern | Salaam Baalak Trust

New Delhi, India | Jun 2024 – Jul 2024

- Mentored children through educational and recreational activities while supporting NGO-led child welfare, learning support, and community outreach programmes.

PROJECTS/RESEARCH

Comparative Simulation of Pursuit Guidance and Proportional Navigation Guidance Laws

MATLAB/Simulink | Jun 2025 – Apr 2026

- Developed and simulated a 3-DOF missile–target engagement model in MATLAB/Simulink to compare Pursuit Guidance and Proportional Navigation based on miss distance, interception time, trajectory response, and guidance command behaviour.

GMAT-Based Hohmann Transfer Orbit Simulation

NASA GMAT | Jun 2025 – Aug 2025

- Simulated a two-impulse Hohmann transfer from a 7000 km circular Earth orbit to a 12000 km circular Earth orbit and analysed orbital parameters, burn timing, and transfer orbit behaviour.

Lunar Rover Hazard-Aware Navigation and Simulation Using MATLAB

MATLAB | Jun 2024 – Aug 2024

- Built a lunar rover navigation simulation for cratered terrain using hazard detection, cost-map generation, A* path planning, Dijkstra comparison, and 2D/3D visualization.

Design and Fabrication of RC Cessna 172 Aircraft Model

XFLR5 | Aug 2024 – Dec 2024

- Designed and fabricated an RC Cessna 172 aircraft model using XFLR5 for airfoil selection, aerodynamic analysis, stability understanding, and aircraft configuration design.

POSITIONS OF RESPONSIBILITY

Group Lead | Team Odyssey

Feb 2026 – Dec 2026

- Led Team Odyssey for the International Space Challenge, Singapore, coordinating research, documentation, and technical presentation work on “Autonomous Orbital Debris Capture, Conversion, and Repair System.”

Group Lead | SRIJAN Social Internship

May 2024 – Aug 2025

- Led a 15-member intern group for social impact initiatives, managing task allocation, planning, coordination, progress reporting, and team communication.

CERTIFICATIONS

International Astronomical Search Collaboration (IASC) — Asteroid Search Campaign

- Analysed Pan-STARRS astronomical images to support observations of near-Earth objects and main-belt asteroids.

ISRO–IIRS Online Training Certifications

- Completed ISRO–IIRS training in AI/ML, geospatial technology, and remote sensing applications.